Liquid Nitrogen

Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Liquid nitrogen; Liquid nitrogen and semen straw

What it is used for

Liquid nitrogen is used in approved livestock breeding and laboratory procedures, including long-term cryogenic storage of semen straws. Safe handling requires trained personnel, suitable cryogenic PPE, proper ventilation, and controlled storage because liquid nitrogen can cause severe cold burns and displace oxygen.

Main parts

- Insulated cryogenic vessel stores liquid nitrogen at very low temperature.
- Lid or neck plug reduces heat entry while allowing safe venting.
- Storage canister holds semen-storage goblets or racks where fitted.
- Canister rack organises semen straws within the vessel.
- Semen straw contains the preserved semen sample.
- Cryogenic gloves and face shield protect against splashes and cold contact.
- Liquid-level measuring device checks the nitrogen level where provided.
- Tongs or approved retrieval tools help handle frozen straws.

Before you start

1. Only trained and authorised personnel should handle liquid nitrogen or retrieve semen straws.
2. Review the vessel instructions, semen-storage procedure, and emergency response plan.
3. Work in a well-ventilated area and ensure oxygen-deficiency monitoring is available where required by the risk assessment.
4. Inspect the cryogenic vessel, lid, canisters, racks, labels, and handling tools for damage.
5. Wear a face shield over safety glasses, loose-fitting cryogenic gloves, protective clothing, and closed footwear as specified by the procedure.
6. Confirm the semen straw identification, storage location, and required retrieval plan before opening the vessel.
7. Keep the vessel upright, stable, and secured against tipping.

How to use

1. Move the cryogenic vessel using the approved handling method and suitable transport equipment.
2. Check the nitrogen level using the specified method without leaning over the vessel opening.
3. Open the vessel carefully and keep the face and body away from escaping cold vapour.
4. Identify the correct canister and semen straw using the storage record.
5. Raise the canister only as far as necessary and keep other stored material submerged as required by the procedure.

6. Use approved cryogenic tongs to retrieve the required straw without touching it with bare skin.
7. Transfer the straw promptly using the approved thawing procedure for that semen product.
8. Return the canister to its correct storage position and close the vessel according to its instructions.
9. Check the nitrogen level after retrieval and replenish only under the approved procedure when required.
10. Record the straw identification, retrieval time, storage status, and any vessel or sample concerns.

⚠ Safety

- Liquid nitrogen can cause severe cold burns and eye injuries; avoid all direct contact.
- Never seal liquid nitrogen in an airtight container because pressure can build up dangerously.
- Nitrogen gas can displace oxygen; never handle or store it in a poorly ventilated area.
- Never lean over the vessel or look directly into its opening.
- Do not use ordinary gloves as a substitute for suitable cryogenic gloves and face protection.
- Do not use a vessel that is damaged, leaking, or showing abnormal frost or pressure behaviour; follow the site emergency plan.

Care & storage

- Keep the vessel upright, clean externally, and in a designated ventilated storage area.
- Inspect the vessel, neck, lid, canisters, and racks regularly for damage or abnormal frost patterns.
- Monitor and record liquid nitrogen levels at the required interval.
- Use only approved methods for filling, moving, and maintaining the vessel.
- Report suspected vacuum failure, unusual nitrogen loss, damaged labels, or sample-storage discrepancies.

Common mistakes to avoid

- Handling liquid nitrogen without suitable cryogenic PPE and training.
- Allowing stored semen straws to warm during unnecessary canister handling.
- Closing liquid nitrogen inside an airtight container.
- Ignoring low nitrogen levels, abnormal frost, or inadequate room ventilation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.